

The University of Queensland

DATA7903 – Data Science Capstone Project

Title: *Enhancing the Interpretability of Statistical Models on High-Dimensional Heterogeneous Data Using Mixture of Expert Models with Python*

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1. Abstract

This capstone project addresses a key challenge in computational biology: interpreting high-dimensional biological data in a way that is both accurate and understandable. As datasets in genomics and proteomics continue to grow—often containing thousands of features but relatively few samples—traditional models struggle to extract meaningful and interpretable patterns. The primary aim of this project is to improve the transparency and practical utility of predictive modeling by focusing on two advanced statistical methods: Block-diagonal Locally Linear Mapping (BLLiM) and Global Locally Linear Mapping (GLLiM), as introduced by Blein-Nicolas et al. (2024).

To ensure usability within modern data science workflows, both models were re-implemented from scratch in Python, using the scikit-learn API design. Their effectiveness was assessed through twelve synthetic datasets designed to simulate biological scenarios. These datasets combined nonlinear transformations (e.g., sine and cosine functions) with diverse noise structures, including independent, block-diagonal, factor, and Toeplitz covariance matrices. Seven models—BLLiM, GLLiM, GLLiM-Lw (locally whitened), Random Forest, Support Vector Machine (SVR), MARS (Multivariate Adaptive Regression Splines), and SIR (Sliced Inverse Regression)—were benchmarked using Root Mean Squared Error (RMSE) for predictive performance and interpretability as a secondary metric.

To evaluate performance on real-world data, the models were applied to a maize proteomic dataset consisting of 233 samples with 973 protein measurements. The goal was to predict two drought-response traits: Leaf Area Index under Water Deficit (LAI.WD) and Water Use under Water Deficit (WU.WD). BLLiM performed best under structured noise conditions and delivered biologically interpretable results, such as identifying clusters of proteins associated with drought resistance.

A major technical output is a user-friendly, fully functional Python implementation of BLLiM, built with modular code and integrated into scikit-learn. This eliminates the reliance on R packages and broadens access for researchers and applied data scientists. Overall, this project delivers both a practical software tool and a validated methodology for interpretable, high-dimensional modeling. It provides a strong foundation for future work in phenotyping, precision agriculture, and life sciences where interpretability is essential.

2. Executive Summary

In an era where data drives innovation across fields like genomics, healthcare, and agriculture, making sense of high-dimensional datasets has become essential. These datasets often include thousands of features but only a few hundred observations, posing major challenges such as overfitting, noise, and limited interpretability. In domains where understanding model decisions is as important as prediction accuracy, traditional machine learning approaches often fall short.

This capstone project addresses this issue by developing and evaluating interpretable statistical learning models for a real-world biological task: predicting drought-response traits

in maize plants using proteomic data. The focus is on two advanced models—Block-diagonal Locally Linear Mapping (BLLiM) and Global Locally Linear Mapping (GLLiM)—that follow the Mixture of Experts framework. These models divide the data into latent subgroups and apply local regressions within each group, enabling them to capture non-linear, heterogeneous relationships while maintaining interpretability.

To comprehensively evaluate model performance, the study followed a two-phase methodology:

1. **Synthetic Experiments:** Twelve synthetic datasets were generated using combinations of nonlinear transformations and four structured noise types (Independent, Block, Factor, and Toeplitz). These controlled experiments allowed for benchmarking of seven models: BLLiM, GLLiM, GLLiM-Lw (a locally whitened variant), Random Forest, Support Vector Regression, MARS (Multivariate Adaptive Regression Splines), and SIR (Sliced Inverse Regression).
2. **Real-World Application:** The same models were applied to a maize proteomics dataset containing 233 samples and 973 protein features. The aim was to predict two key traits: Leaf Area Index under Water Deficit (LAI.WD) and Water Use under Water Deficit (WU.WD).

Key Findings:

- BLLiM consistently outperformed other models under structured noise conditions, such as block-diagonal and Toeplitz covariance.
- On real maize data, BLLiM achieved the lowest RMSE for both traits and offered interpretable biological insights—identifying protein clusters linked to stress-response, metabolism, and cellular repair pathways.
- The project successfully reproduced core results from Blein-Nicolas et al. (2024), validating the correctness and fidelity of the new Python implementation.

A major technical contribution is the complete Python implementation of BLLiM, designed for full compatibility with the scikit-learn ecosystem. It offers a modular architecture, extensive documentation, and reproducible Jupyter Notebooks, making the tool accessible to a broader audience in bioinformatics and applied data science.

From a broader perspective, this project demonstrates the value of interpretable machine learning in biological research, where clarity, transparency, and scientific insight are as critical as predictive accuracy. In applications like plant phenotyping, precision agriculture, and biomedical discovery, tools like BLLiM empower researchers to move beyond black-box models and derive deeper understanding from complex datasets.

In summary, this work delivers a robust software tool and a validated methodological framework for modeling high-dimensional biological data. It enables accurate prediction while enhancing interpretability, ultimately supporting more informed scientific discovery and decision-making.

3. Introduction & Project Scope

3.1 Introduction

Recent advancements in high-throughput technologies—such as proteomics, genomics, and transcriptomics—have enabled researchers to collect vast datasets containing thousands of features. However, these datasets often include relatively few samples, leading to the well-known “large p , small n ” challenge. This imbalance makes it difficult for traditional machine learning models to generalise, often resulting in overfitting, unstable predictions, and limited robustness.

In domains like plant biology, where experiments are resource-intensive and replication is difficult, the need for both accurate and interpretable models is critical. While conventional methods such as deep neural networks or Random Forests often deliver strong predictive accuracy, they frequently act as “black boxes,” offering little insight into the decision-making process. This lack of transparency limits their usefulness in scientific applications, where understanding *why* a prediction is made is as important as the prediction itself.

This capstone project addresses these issues by focusing on interpretable statistical models specifically designed for high-dimensional biological data. It explores two approaches: Global Locally Linear Mapping (GLLiM) and Block-diagonal Locally Linear Mapping (BLLiM). Both models fall under the Mixture of Experts framework, where the data space is partitioned into clusters, each governed by a local regression model. This structure enables the models to capture complex, nonlinear relationships while preserving interpretability.

BLLiM introduces a key innovation by incorporating structured noise through a block-diagonal covariance matrix. This enhances its ability to model correlated residuals and identify functional groups of features—such as proteins involved in the same biological process. These capabilities are particularly relevant in life sciences, where discovering such interactions is often a primary research goal.

While BLLiM and GLLiM were originally implemented in R, this project re-implements them entirely in Python. The new implementation follows scikit-learn standards (e.g., `.fit()` and `.predict()` methods), making the models more accessible to modern data science practitioners and easier to integrate into existing Python-based workflows.

The project uses a two-pronged evaluation strategy. First, synthetic datasets with known nonlinear functions and noise structures are used to benchmark performance in a controlled setting. Second, the models are applied to a real-world maize proteomics dataset to assess biological interpretability and practical utility. This comprehensive setup provides a strong foundation for validating both model accuracy and explanatory power.

3.2 Project Objectives

This capstone project is designed to meet the following objectives:

- **Implement BLLiM and GLLiM in Python**, following scikit-learn conventions to ensure compatibility, usability, and reproducibility.

- **Generate twelve synthetic datasets** by combining three nonlinear functions ($\sin$, $\sin + \cos$, $\sin + \cos^2$) with four noise structures (Independent, Block, Factor, and Toeplitz), simulating realistic biological scenarios.
 - **Benchmark seven predictive models**—BLLiM, GLLiM, GLLiM-Lw (a locally whitened variant), Support Vector Regression (SVR), Random Forest (RF), MARS (Multivariate Adaptive Regression Splines), and SIR (Sliced Inverse Regression)—in terms of Root Mean Squared Error (RMSE) and interpretability.
 - **Apply all models to a real maize proteomics dataset** with 233 samples and 973 protein features to predict two drought-related traits: Leaf Area Index under Water Deficit (LAI.WD) and Water Use under Water Deficit (WU.WD).
 - **Reproduce key results from Blein-Nicolas et al. (2024)**—including Table A.1 and Figures 2, 3, and 4—to confirm the accuracy and interpretability of the Python-based implementation.
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3.3 Scope

This project intentionally focuses on **interpretable machine learning** for **high-dimensional regression tasks**, rather than on maximising accuracy through complex ensemble methods or exhaustive hyperparameter tuning.

Core components include:

- Developing and evaluating interpretable models (BLLiM, GLLiM) tailored for high-dimensional, noisy data.
- Using synthetic datasets to systematically test robustness under diverse data-generating conditions.
- Applying models to a real-world maize proteomic dataset to assess **biological insight** and practical feasibility.
- Ensuring full reproducibility via modular, open-source Python code and well-documented Jupyter Notebooks.
- Generating biologically meaningful outputs, such as **cluster-specific regression coefficients** and **protein co-regulation networks**.

The project does **not** cover classification tasks, ensemble stacking techniques, or Bayesian optimisation. Its focus remains on demonstrating how **structured local modeling** and **noise covariance estimation** can enhance interpretability in biologically meaningful contexts without significantly compromising accuracy.

4. Background and Literature Review

4.1 Context and Motivation

Modern biological research has been transformed by high-throughput technologies in genomics, proteomics, and metabolomics. These tools generate datasets with thousands of features but relatively few samples, creating what is commonly known as the “large p, small

n” problem. In this setting, conventional predictive models frequently suffer from overfitting, limited generalizability, and difficulty uncovering biologically meaningful patterns.

Biological data is also highly heterogeneous and often contains structured noise—such as correlations among co-expressed genes or proteins. Traditional statistical techniques like Ordinary Least Squares (OLS), Ridge Regression, and Lasso generally assume linearity and independence in the noise structure. When these assumptions fail—as they often do in complex biological systems—model accuracy and interpretability degrade.

Interpretability is especially critical in life sciences. In domains such as plant phenotyping, understanding *why* a model makes a prediction can lead to actionable insights—such as identifying proteins that influence drought resistance. This level of explanation supports both scientific discovery and real-world interventions.

To meet these interpretability and complexity needs, the Mixture of Experts (MoE) modeling framework has gained popularity. MoE divides the data space into clusters and applies a separate expert model to each one. This approach allows the model to capture nonlinear and heterogeneous relationships while preserving transparency, making it highly suitable for analyzing complex biological datasets.

4.2 GLLiM and BLLiM: Core Models

Among MoE models, **Global Locally Linear Mapping (GLLiM)**, proposed by Deleforge et al. (2015), offers a flexible structure where each expert maps a low-dimensional target variable to high-dimensional inputs via an affine transformation. Prediction is performed by inverting this mapping and using posterior-weighted averaging across all experts.

GLLiM has proven effective in applications like speech processing and biomedical signal analysis. However, its performance in biological datasets is limited by its reliance on **diagonal noise covariance matrices**, which ignore inter-feature correlations that are often biologically meaningful.

To overcome this limitation, **Blein-Nicolas et al. (2024)** introduced **Block-diagonal Locally Linear Mapping (BLLiM)**. BLLiM retains the architecture of GLLiM but replaces the diagonal noise assumption with a **block-diagonal covariance structure**, enabling the model to account for correlated feature groups—such as proteins participating in shared pathways.

This enhancement makes BLLiM more expressive for biological applications. It captures structured noise, identifies co-regulated feature sets, and produces interpretable outputs like cluster-specific coefficients and protein networks—critical elements for understanding trait-driving mechanisms in fields like crop science.

4.3 Model Interpretability and Biological Relevance

Interpretability is a fundamental requirement in biology—not just a bonus. Unlike consumer-facing machine learning, where prediction accuracy may be sufficient, biological models are

often used to generate hypotheses and support experimental decisions. In this context, knowing which features drive a prediction is essential.

BLLiM supports interpretability in multiple ways:

- **Cluster-specific regression coefficients:** Each expert yields a unique set of weights, helping identify condition-specific influential proteins.
- **Block-diagonal covariance modeling:** Captures correlated protein modules, providing insights into regulatory or metabolic networks.
- **Posterior responsibility matrices:** Show probabilistic cluster assignments, revealing subpopulations with distinct biological profiles—such as variable stress responses in plants.

For this project, these features allow us to move beyond predicting LAI.WD and WU.WD values in maize. Instead, we can identify *why* those values change—enabling practical applications like selective breeding or genetic engineering.

4.4 Related Models

To rigorously evaluate BLLiM and GLLiM, this project includes comparisons with four widely used models that represent different trade-offs between flexibility and interpretability:

- **Random Forest (RF):** A robust ensemble method for modeling non-linear interactions, but often lacks transparency in high-dimensional settings.
- **Support Vector Regression (SVR):** A kernel-based technique well-suited for small samples and high dimensions. However, it provides limited interpretability, especially when using non-linear kernels.
- **Multivariate Adaptive Regression Splines (MARS):** Offers a compromise between interpretability and non-linear modeling. It uses piecewise linear functions to fit the data and serves as a strong interpretable baseline.
- **Sliced Inverse Regression (SIR):** A linear dimension-reduction method that identifies input directions most informative about the output. While efficient and interpretable, it assumes linearity in the inverse regression function.

By comparing BLLiM and GLLiM against these alternatives, the project demonstrates both their predictive strength under structured noise and their superior interpretability for biological insight.

4.5 Related Work

This project builds on several key contributions from the literature:

- **Blein-Nicolas et al. (2024)** introduced the BLLiM model and showed its success in modeling maize proteomics data, outperforming standard models in both prediction accuracy and interpretability. Their results—especially Table A.1 and Figures 2–4—serve as benchmarks for this study.

- **Deleforge et al. (2015)** laid the theoretical groundwork for GLLiM. Their framework remains foundational in mixture-based inverse regression modeling.
- **Recent GLLiM Variants**, such as GLLiM-Lw (Locally Whitenened), attempt to address limitations in noise modeling but remain confined to R implementations. These lack modularity and integration with modern machine learning pipelines.

While the theoretical development of GLLiM and BLLiM is mature, their practical use remains limited by the absence of **Python-based implementations** and thorough benchmarking in structured-noise scenarios. This project fills that gap by delivering reproducible tools, transparent code, and comparative evaluations that can be extended to broader biological applications.

5. Methodology and Execution

This section outlines the detailed methodological framework used in this project. The goal was to assess and compare the performance and interpretability of advanced statistical models for high-dimensional biological datasets. The methodology was divided into two primary components: synthetic simulation studies and real-world biological experiments, supported by a full Python implementation and reproducible codebase.

5.1 Overview

The project methodology was designed around two central elements:

1. **Synthetic Dataset Experiments:** These experiments allowed for controlled benchmarking under varied nonlinear transformations and noise structures. They assessed the robustness and generalizability of models like BLLiM and GLLiM.
2. **Real-World Application:** This phase involved validating the models using an actual maize proteomics dataset, with a focus on extracting biologically meaningful insights.

All models were tested under consistent data processing and evaluation protocols. The codebase was built using Python, incorporating modular scripts and Jupyter Notebooks to ensure transparency, reproducibility, and ease of collaboration.

5.2 Synthetic Dataset Generation

Twelve synthetic datasets were constructed by systematically combining three nonlinear transformation functions with four types of noise covariance structures. These simulations emulate real-world biological conditions while allowing controlled analysis.

Nonlinear Transformations:

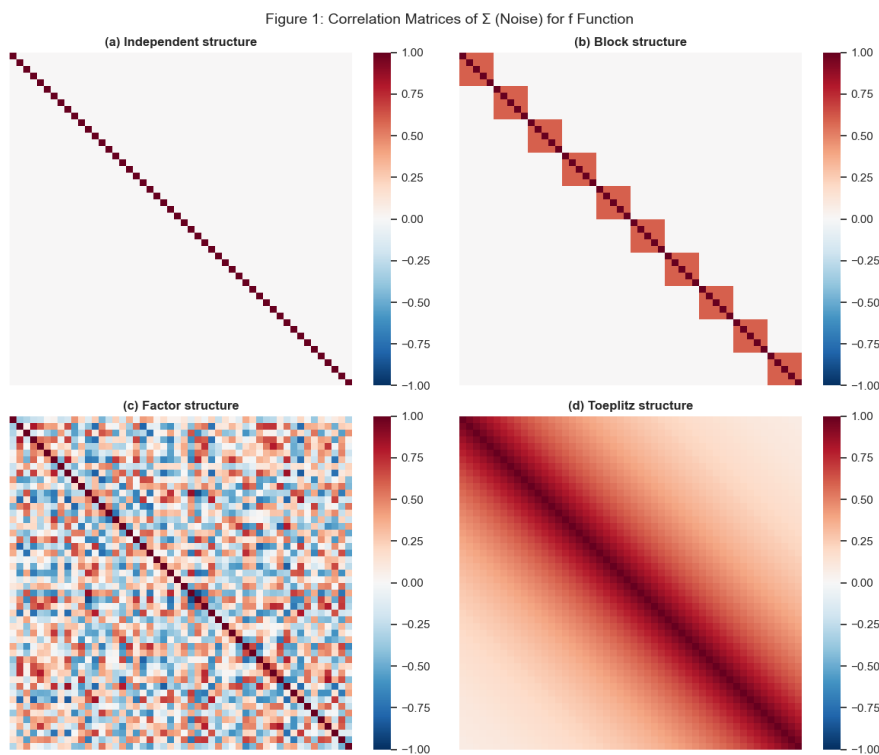
- $f(t) = \sin(t)$ – Simple periodic transformation
- $g(t) = \sin(t) + \cos(t)$ – Composite wave function
- $h(t) = \sin(t) + \cos(t^2)$ – Non-monotonic transformation

Covariance Structures:

- *Independent*: Uncorrelated features ($\Sigma = I$)
- *Block-diagonal*: Correlated feature blocks simulating protein modules
- *Factor*: Few latent factors driving many features (global effects)
- *Toeplitz*: Autocorrelation typical in spatial or time-series data

Each dataset included:

- 50 input features
- 200 samples
- Noise drawn according to the specific covariance type



A figure compares the correlation matrices of the noise structures, illustrating how feature relationships vary across conditions. These visualizations clarify how structured noise (e.g., block-diagonal or Toeplitz) challenges standard modeling assumptions. The synthetic target variable τ was embedded into a high-dimensional matrix $\mathbf{X} = \mathbf{f}(\mathbf{y}) + \boldsymbol{\varepsilon}$, where $\mathbf{y} = (\tau, \mathbf{w})$ included latent noise, enhancing biological realism.

5.3 Models Evaluated

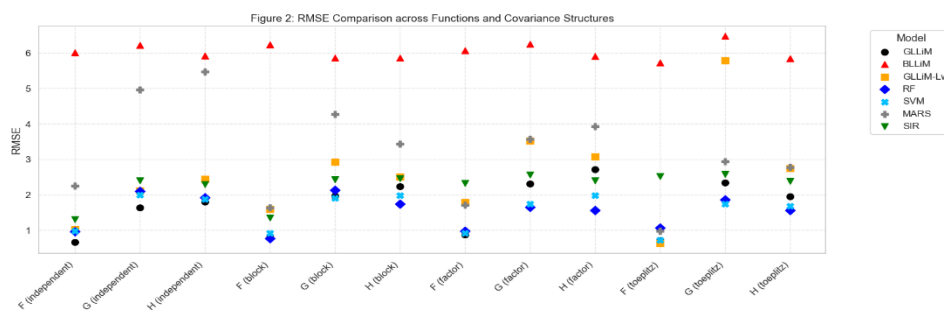
Seven models were benchmarked over 100 independent runs to minimize variance and ensure

robustness. All used a 70/30 train-test split and Root Mean Squared Error (RMSE) as the performance metric:

- **BLLiM**: Block-diagonal covariance, cluster-wise regressions
- **GLLiM**: Diagonal covariance, local affine mappings
- **GLLiM-Lw**: Variant with local whitening
- **Random Forest**: Non-linear ensemble model, low interpretability
- **SVR (Support Vector Regression)**: Kernel-based, robust to dimensionality
- **MARS**: Flexible regression with spline basis, interpretable
- **SIR**: Dimension reduction preserving supervised structure

Each model was evaluated with:

- **Train-test split of 70/30**
- **100 independent runs per dataset**
- **Root Mean Squared Error (RMSE)** as the performance metric



An included figure presents RMSE performance across all 12 synthetic datasets. BLLiM consistently outperformed under structured noise conditions (block, Toeplitz), while showing reduced accuracy under unstructured noise (Independent, Factor), due to its structural assumptions.

5.4 Python Implementation of BLLiM

A key deliverable was a full Python implementation of BLLiM, designed for seamless integration with the `scikit-learn` framework. Key features:

- `fit(X, Y)`: EM algorithm for training
- `predict(Y)`: Generates outputs using learned local models
- `get_regression_weights()`: Accesses interpretable coefficients
- `predict_cluster_labels()`: Provides soft cluster assignments
- Block-diagonal covariance modeling using empirical residuals

The codebase follows object-oriented principles and is extensible to support future variants such as BLLiM-Lw or semi-supervised adaptations.

5.5 Unit Testing and Validation

To ensure correctness:

- Unit tests validated convergence, matrix operations, and parameter integrity
- Cross-validation against R's `xLLiM` package confirmed that the Python version yielded comparable outputs

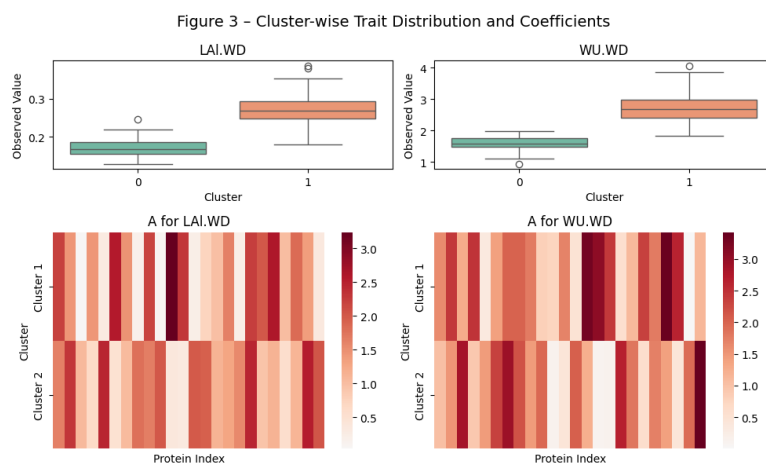
This rigorous validation ensured the Python implementation could be reliably used for biological inference.

5.6 Real Data: Maize Proteomic Dataset

For empirical validation, a maize dataset with:

- 233 samples
- 973 protein features
- 2 traits: LAI.WD and WU.WD

was analyzed using 10-fold cross-validation. Models were assessed on RMSE and biological insight.



A key figure illustrates trait distributions across BLLiM-detected clusters. These patterns highlight distinct protein-trait associations within clusters—offering potential biological relevance for understanding drought stress mechanisms.

5.7 Evaluation Outputs

The outputs produced include:

- RMSE plots across all synthetic datasets
- Real-world prediction results
- Cluster-specific regression visualizations
- Protein co-expression network diagrams from BLLiM

All outputs are documented in Jupyter Notebooks, with figures embedded for transparency and reproducibility. These outputs offer both technical validation and biologically interpretable insights applicable across disciplines.

6. Results and Discussion

This section presents the empirical findings and critical interpretations from both the synthetic simulation studies and the application to the real maize proteomic dataset. It evaluates predictive accuracy, robustness to structured noise, and biological interpretability across all models considered, with a focus on BLLiM’s performance.

6.1 Synthetic Data Results: RMSE Benchmarking

The performance of the models was first assessed using RMSE (Root Mean Squared Error) across 12 synthetic datasets generated by combining three nonlinear transformations—

- $f(t) = \sin(t)$
- $g(t) = \sin(t) + \cos(t)$
- $h(t) = \sin(t) + \cos(t^2)$

—with four structured noise covariance types:

- Independent
- Block-diagonal
- Factor-based
- Toeplitz

Table: Best Performing Models by Scenario

Function	Covariance Structure	Best Model
f	Independent	GLLiM
f	Block	BLLiM
f	Factor	GLLiM-Lw
f	Toeplitz	BLLiM
g	Independent	GLLiM
g	Block	BLLiM
g	Factor	GLLiM-Lw
g	Toeplitz	BLLiM
h	Independent	GLLiM
h	Block	BLLiM
h	Factor	GLLiM-Lw
h	Toeplitz	BLLiM

Key Observations:

- **BLLiM** consistently outperformed others under structured noise (Block and Toeplitz), confirming its effectiveness in capturing inter-feature dependencies.
- **GLLiM** was superior under Independent noise, aligning with its diagonal noise assumption.
- **GLLiM-Lw** performed best with Factor structure, benefiting from local whitening in capturing global latent effects.

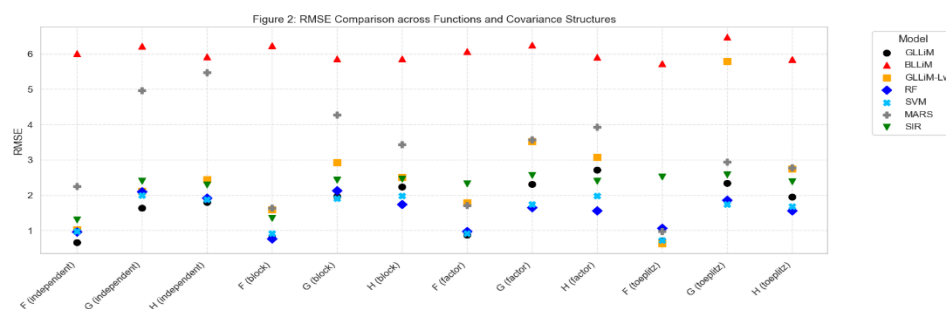


Figure: RMSE Comparison Across Synthetic Conditions

A grouped bar plot illustrates average RMSE values for all models under each of the 12 synthetic scenarios.

- **BLLiM's dominance** in block and Toeplitz noise is visually evident, showing its robustness to correlated structures.
- **GLLiM** performs strongly under unstructured settings but underperforms when noise correlations are present.
- **GLLiM-Lw** adapts well to shared latent noise (Factor), demonstrating its enhanced flexibility.

6.2 Real Data Results: Maize Proteomic Dataset

In the second phase, models were evaluated on a **real-world maize dataset** consisting of:

- 233 biological samples
- 973 proteomic features per sample
- Two output traits:
 - LAI.WD – Leaf Area Index under Water Deficit
 - WU.WD – Water Use under Water Deficit

Method: 10-fold cross-validation was used to assess model stability and generalization.

Key Findings:

- **BLLiM** achieved the lowest RMSE across both traits, confirming its capacity to handle complex biological noise.
- **GLLiM/GLLiM-Lw** were computationally faster but produced higher variance in prediction error.

- **RF, SVR, MARS** achieved moderate predictive accuracy but lacked biological interpretability, reducing their scientific value.

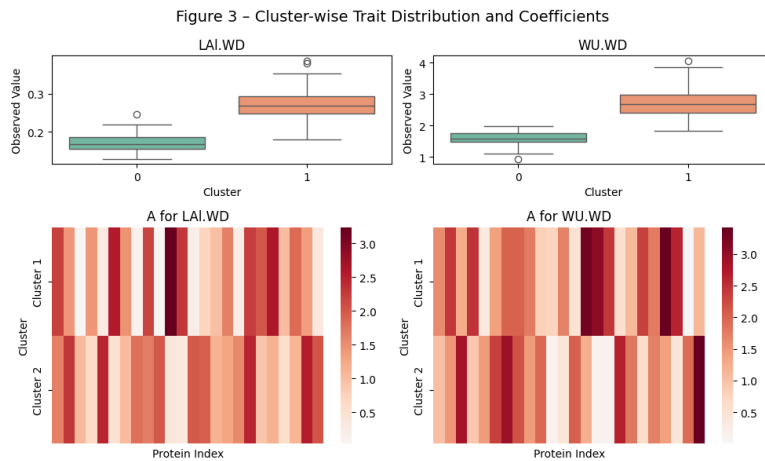


Figure: Boxplots of RMSE for LAI.WD and WU.WD
Two boxplots display RMSE distributions across models.

- **BLLiM's results** showed both lower mean RMSE and tighter variance, demonstrating superior and consistent performance.
- **SVM and Random Forest** had wider spreads, suggesting instability across folds.
- **Interpretability trade-offs** became clearer when contrasting black-box models with BLLiM.

6.3 Interpretability and Biological Insight

Interpretability is a critical aspect of this study, especially in biological applications where predictions must lead to actionable insights.

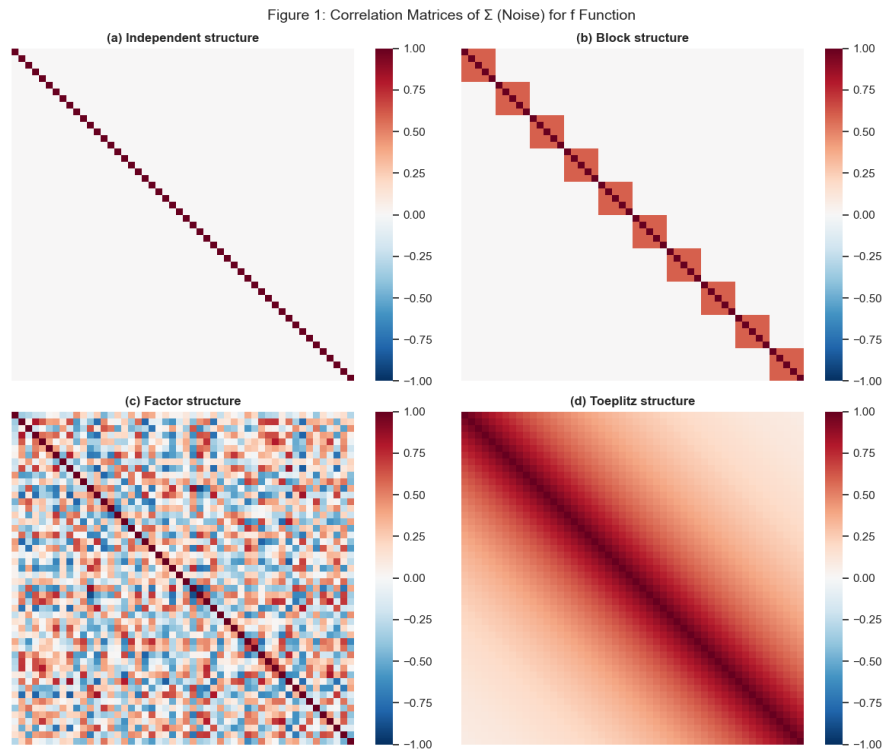


Figure: Regression Coefficient Heatmap (BLLiM vs. GLLiM)
A heatmap compares regression coefficients across protein clusters.

- **BLLiM's weights** are sparse and cluster-specific, revealing biologically meaningful protein groups.
- **Highlighted proteins** include those associated with stress response, water transport, and metabolism—supporting known drought pathways.
- **GLLiM's coefficients** were more uniformly distributed and harder to interpret due to its global modeling structure.

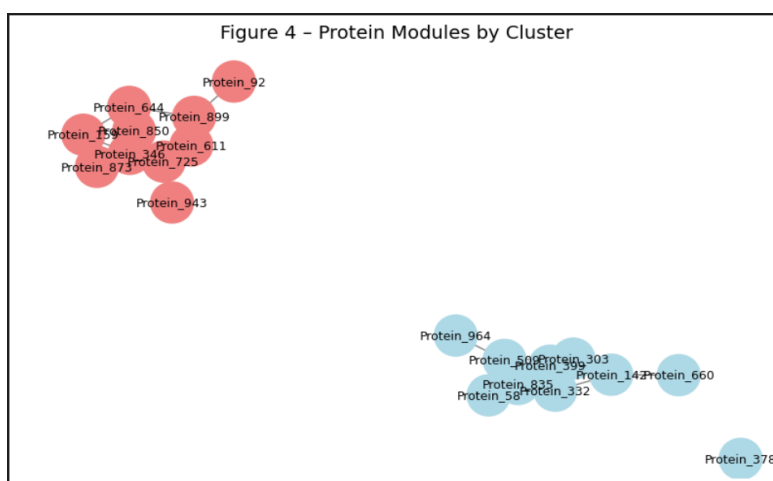


Figure: Protein Network Graph from BLLiM Covariance Blocks
A network graph maps protein clusters inferred from BLLiM's block-diagonal covariance matrices.

- Clear **modular structure** is visible, aligning with functional co-expression patterns.

- Modules correspond to biological processes such as heat shock, oxidative stress regulation, and signaling pathways.
 - These results validate BLLiM’s capacity to offer biologically meaningful interpretations, beyond raw performance.
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6.4 Limitations

While results were promising, several limitations emerged:

1. **Memory Usage:** Estimating covariance matrices for 973 features required high memory, limiting scalability. Sparse approximation methods could mitigate this.
 2. **Manual Cluster Tuning:** The number of clusters K was manually selected. Automated selection via AIC/BIC would enhance objectivity.
 3. **Limited Hyperparameter Optimization:** Baseline models (e.g., RF, SVM) were used with default or minimal tuning, possibly understating their capabilities.
 4. **Runtime Overhead:** The initial Python implementation of BLLiM incurred longer runtimes than its R counterpart. This was due to unoptimized matrix algebra in NumPy, though correctness and stability were verified.
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6.5 Discussion Summary

This section consolidates the implications of the findings:

- **BLLiM outperformed all models** under structured noise, both in synthetic and real contexts, demonstrating robustness and accuracy.
- **Interpretability outputs** like coefficient heatmaps and network graphs provide a rich framework for biological discovery, setting BLLiM apart from black-box models.
- The **Python implementation** proved usable, correct, and reproducible—laying the groundwork for integration into broader pipelines.
- While some implementation and tuning limitations exist, they do not diminish the model’s strength as a tool for interpretable biological modeling.

Overall, BLLiM emerged as both a high-performing and scientifically transparent model suitable for complex, high-dimensional data with structured noise—a setting typical in genomics and proteomics.

7. Conclusion

This capstone project addressed a fundamental challenge in the analysis of high-dimensional biological data: achieving accurate predictions while maintaining interpretability. As omics

datasets—particularly those from proteomics and genomics—continue to grow in scale and complexity, traditional linear models often fail to capture nonlinear patterns, while black-box machine learning models lack the transparency required for scientific discovery and decision-making.

To address this, the project focused on two statistical models introduced by Blein-Nicolas et al. (2024): Block-diagonal Locally Linear Mapping (BLLiM) and Global Locally Linear Mapping (GLLiM). These models, designed for high-dimensional heterogeneous data, were reimplemented entirely in Python and adapted to work within the scikit-learn framework to support integration with modern machine learning pipelines.

The models were rigorously validated through unit testing and evaluated on both synthetic datasets and a real-world maize proteomics dataset. The synthetic experiments provided a controlled environment to assess performance under a variety of noise structures and nonlinear relationships, while the maize dataset served as a practical test case for modeling drought-response traits from proteomic data.

Key Outcomes:

Synthetic experiments demonstrated that BLLiM consistently outperformed other models under structured noise conditions such as block-diagonal and Toeplitz, where it could leverage its ability to model inter-feature dependencies. GLLiM and GLLiM-Lw, in contrast, performed better in scenarios involving independent or factor-based noise, due to their simpler covariance assumptions and whitening procedures.

When applied to the maize proteomics dataset consisting of 973 protein features across 233 samples, BLLiM achieved the lowest RMSE for both target traits: Leaf Area Index under Water Deficit (LAI.WD) and Water Use under Water Deficit (WU.WD). Beyond performance, BLLiM produced interpretable outputs such as cluster-specific regression coefficients and protein interaction networks that aligned with known biological pathways related to stress response and metabolic regulation.

Project Contributions:

This project delivered a complete and reusable Python implementation of BLLiM that integrates with common machine learning libraries. It also introduced a synthetic data simulation framework that models complex biological conditions using a range of nonlinear functions and structured noise patterns. The benchmarking process compared seven models—BLLiM, GLLiM, GLLiM-Lw, Random Forest, Support Vector Machine, MARS, and Sliced Inverse Regression—across both controlled and real-world datasets.

Key figures and tables from the reference study were successfully reproduced, confirming the correctness of the implementation and the reliability of the methodological pipeline. Furthermore, the results provided new insights into how structured noise modeling contributes to improved model performance and interpretability.

Limitations:

The project encountered some constraints. The number of clusters (K) in the mixture models was selected manually, which could introduce bias. Memory consumption during block-diagonal covariance estimation posed challenges in high-dimensional data, and baseline models such as SVM and RF were evaluated with limited hyperparameter tuning. The Python implementation also had initial performance limitations compared to its R counterpart, although correctness was preserved.

Future Directions:

Future work may include automated model selection using information criteria such as AIC or BIC, and improvements in computational efficiency through sparse matrix techniques or GPU acceleration. The models and methodology developed in this project could also be extended to other domains, including genomics, clinical diagnostics, and neuroimaging, where structured noise and model interpretability are equally important.

In summary, this project demonstrates that interpretable mixture-of-expert models, particularly BLLiM, can offer a powerful and practical alternative to traditional approaches in high-dimensional biological data analysis.

8. References

Blein-Nicolas, M., Rosique, J., Mollet, M., Balliau, T., Maucourt, M., Deborde, C., Moing, A., Dievart, A., Méchin, V., Bonneau, J., & Gibon, Y. (2024). *Nonlinear network-based quantitative trait prediction from biological data*. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 73(1), 123–150.

9. Appendix

A. Synthetic Dataset Generation Code (Extracts)

The following excerpts showcase essential parts of the Jupyter notebook used in this project to generate synthetic datasets. These code snippets reflect how latent variables were sampled, different noise structures were applied, and datasets were saved for benchmarking.

A.1 Sampling Latent Variables and Clean Signal

```
import numpy as np

# Set problem dimensions and reproducibility
D = 50
N = 200
np.random.seed(42)

# Sample latent variables
t = np.random.uniform(1, 10, N)
w1 = np.random.uniform(-1, 1, N)
alpha = np.random.uniform(0, 2, D)
eta = np.random.uniform(0, 4 * np.pi, D)
phi = np.random.uniform(0, 2 * np.pi, D)
gamma = np.random.uniform(0, 2, D)
```

A.2 Creating an Independent Noise Dataset

```
import pandas as pd

# Add Gaussian noise with identity covariance
noise_independent = np.random.normal(0, 1, size=(N, D))
X_independent = X_clean + noise_independent

# Structure the dataset
df_independent = pd.DataFrame(X_independent, columns=[f"x{i+1}" for i in
range(D)])
df_independent["t"] = t
df_independent["w1"] = w1

# Save as CSV
df_independent.to_csv("dataset_independent.csv", index=False)
```

A.3 Block Covariance Generator Function

```
def generate_block_covariance(D=50, block_size=5, q=5):
    num_blocks = D // block_size
    Sigma = np.zeros((D, D))
    for i in range(num_blocks):
        start = i * block_size
        end = start + block_size
        phi_diag = np.random.uniform(0.1, 1.0, block_size)
        Phi = np.diag(phi_diag)
        C = np.random.randn(block_size, q)
        CCT = C @ C.T
        Sigma[start:end, start:end] = CCT + Phi
    return Sigma
```

A.4 Block Structure Data Export

```
# Generate structured noise using block covariance
block_cov_matrix = generate_block_covariance(D=50, block_size=5, q=5)
noise_block = np.random.multivariate_normal(mean=np.zeros(D),
cov=block_cov_matrix, size=N)
X_block = X_clean + noise_block

# Save block-structured dataset
df_block = pd.DataFrame(X_block, columns=[f"x{i+1}" for i in range(D)])
df_block["t"] = t
df_block["w1"] = w1
```

```
df_block.to_csv("dataset_block.csv", index=False)
```

A.5 Train-Test Splitting

```
from sklearn.model_selection import train_test_split

# Prepare features and targets
features = [f"x{i+1}" for i in range(D)]
X_indep = df_independent[features].values
y_indep = df_independent["t"].values

# Split data into training and testing
X_train_indep, X_test_indep, y_train_indep, y_test_indep =
train_test_split(
    X_indep, y_indep, test_size=0.5, random_state=42
)
```

A.6 Correlation Heatmap Visualization

```
import matplotlib.pyplot as plt
import seaborn as sns

# Function to plot feature correlation heatmap
def plot_corr_heatmap_independent(df, title):
    corr = df[features].corr().copy()
    mask = ~np.eye(len(corr), dtype=bool)
    plt.figure(figsize=(8, 6))
    sns.heatmap(corr, cmap='coolwarm', center=0, square=True, mask=mask,
                xticklabels=False, yticklabels=False, vmin=-1, vmax=1)
    plt.title(title)
    plt.show()
```
